CANDY
DIALOGUE ENGINE

Bust (VN) Scene Customization

1.1.0

Bust scenes are used for visual novels. They contain child nodes that are used for displaying "bust nodes", which depict character busts.

Bust scenes are intended to be customized in the following ways by users:

- Size and position of the bust nodes on screen.
- Bust node types (TextureRect, Sprite2D/3D, etc).
- Name of bust nodes.
- Adding speech bubbles to bust nodes.
- Adding AnimationPlayer nodes to bust nodes.
- Adding AnimationPlayer nodes to the bust scene.

Bust scenes are meant to be swapped at runtime, as needed. The §VN_Scene command is used for that purpose. This allows you to easily switch VN layouts for different parts of your game.

Nodes

Root

The root node can have any desired name.

Required Nodes

The following nodes are required:

Bust Container

- Can be any node type.
- Acts as a parent for bust nodes.
- Must not have any other children.

- Script pointer: @export var **busts_container**

Bust Nodes

- Display character busts.
- TextureRect, TextureButton, Sprite2D/3D, AnimatedSprite2D/3D or VideoStreamPlayer.
- Can have any name.
- The scene can feature any number of bust nodes.
- Must be direct children of the Busts Container.
- No script pointer.

Optional Nodes

Bust Speech Bubbles

- Used for displaying spoken text in a comic-book style speech bubble.
- Must be a speech bubble scene (see the **Speech Bubbles** guide and the **Speech Bubbles Customization** guide).
- Must be named "SpeechBubble".
- Must be a direct child of the specific bust node it displays speech for.
- No script pointer.

Bust Animation Players

- Used for playing animations on a specific bust node.
- Must be AnimationPlayer.
- Must be named the same as the value of the **jml_player_name** variable (see below).
- Must be a direct child of the specific bust node it plays animations on.
- No script pointer.

Bust Outline Nodes

- Used for displaying an outline around a bust, for the purpose of highlighting the speaking character.
- TextureRect, TextureButton, Sprite2D/3D, AnimatedSprite2D/3D or VideoStreamPlayer.
- Must be a child of a bust node.

- Must be named "Outline" (customizable in the **highlight_speaker()** and **end_highlight_speaker()** functions)
- Requires writing custom code in the **highlight_speaker()** and **end_highlight_speaker()** functions. Won't work out-of-the-box.

Highlight Animation Players

- Optionally used for playing animations on the bust of the speaking character, for the purpose of highlighting.
- Must be AnimationPlayer.
- Can be given any name.
- Must NOT be a child of any bust node.
- Each such AnimationPlayer requires a script pointer. The name of the script pointer can be customized.
- Script pointer: @export var **highlight_player_1**
- Read further or see the **Visual Novels** guide for more details about speaker highlighting.

Script

Base

The script **Bust_Scene.gd** should be attached to the scene's root node.

Variables

Required:

default_image_extension

Indicates the extension to use by default for bust files, if no extension is provided with the file name, when the bust node is a TextureRect or TextureButton.

default_sprite_extension

Indicates the extension to use by default for bust files, if no extension is provided with the file name, when the bust node is a Sprite2D or Sprite3D.

default_animated_sprite_extension

Indicates the extension to use by default for bust files, if no extension is provided with the file name, when the bust node is an AnimatedSprite2D or AnimatedSprite3D.

default_video_extension

Indicates the extension to use by default for bust files, if no extension is provided with the file name, when the bust node is a VideoStreamPlayer.

sprite_fps

The FPS (frames per second) value to use when playing animated busts with Sprite2D/3D or AnimatedSprite2D/3D nodes.

bust_sizes

The size you want to display each Sprite2D/3D or AnimatedSprite2D/3D bust nodes at. These nodes can't be assigned a size directly, so must be scaled automatically through code.

vn_join_effect / vn_move_from_effect / vn_move_to_effect / vn_leave_effect

Data for the JML effects to play when a character is assigned to a bust or unassigned from a bust.

See the **Visual Novels** guide for details.

speaker_highlight

A dictionary for enabling and configuring speaker highlighting effects. See the **Visual Novels** guide for details.

billboard_on / billboard_off

Used only for Sprite3D and AnimatedSprite3D nodes. Indicates the billboard mode to switch to when billboard mode is toggled on or off on a bust node, as part of the speaker highlighting options.

See the **Visual Novels** guide for details.

bust_node_data

Used by the code to store data about busts.

default_z
The default z_index value for the bust scene. Should be lower than UI elements for displaying dialogues, such as the dialogue box.

caller

The dialogue engine instance that is using the bust scene.

Functions

highlight_speaker()

This function contains customizable logic for highlighting the bust of the speaker character (i.e. the character speaking a dialogue line at any time).

The function is called automatically by the dialogue engine.

The code is theoretically fully customizable, but we recommend expanding upon the default template.

The default template uses a match statement to select different highlighting logic depending on dialogue mode. For readability, only the "Box" dialogue mode has any code by default, but you can just copy/paste that code to the other dialogue modes.

The default template also provides logic for the different highlight options. These options can be enabled/disabled or further configured in the **speaker_highlight** dictionary.

Take a few minutes to examine the code, you should find it pretty straightforward.

Note that the provided code for the 'Outline' highlight option is just a stub:

The code only features an option to hide the 'real' speaker bust by making it transparent, and does not actually feature logic to display an outline.

If you want to display an outline around the speaker, you'll need to write your own logic. A few ideas:

1. Hide the real bust image, and display an alternate bust image with an outline drawn around it. To help the code find the correct outline image file automatically, name outline files the same as bust files, with an added suffix of your choice (e.g. Angry.png → Angry_OL.png)
2. Don't hide the real bust image; under the real image, display a copy of the bust image, increase its scale slightly, and modulate it to a single flat color.
3. Overlay an outline node and the bust node, and write code or a shader to draw an outline in the outline node.

For what it's worth, we're considering providing default code for options 1 and 2 in the future. At this time, we're trying to strike a balance between feature richness and code clarity. We also await user feedback to determine which outline options should be added by default, based on user preferences and common practices.

Overall, **highlight_speaker()** is very powerful and modular, enabling you to create virtually any speaker highlighting effect you may imagine. However, logic beyond the provided default must be programmed by yourself, the complexity of which depends on the effects you are attempting to achieve.

end_highlight_speaker()

This function is the reverse of **highlight_speaker()**: it basically reverts what **highlight_speaker()** did.

If you add or modify logic in **highlight_speaker()**, you must ensure that logic is mirrored (reverted) in **end_highlight_speaker()**.

_on_play_jml()

This function handles the logic for playing JML effects.

You can add custom methods under the 'match jml_method:' statement.

You may also want to move the position of the function call in other functions, to time it differently.

Other functions

All other functions in the script are required. Editing these functions is not recommended unless you have a good reason to do so, and you know what you're doing.